Exercise 13 - Describing Functions

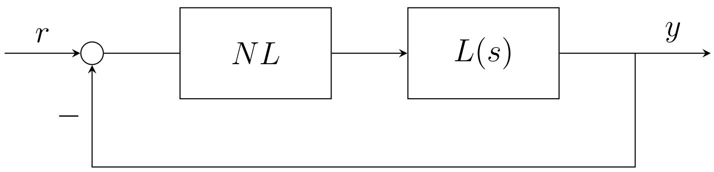
1 Describing Functions

As discussed last week, in this course we mainly focused on linear systems. However, many real-world systems exhibit nonlinear behavior.
Until now, we have handled this nonlinerities by linearizing the system around some equilibrium point. This approach however, cannot predict all phenomena that can arise in nonlinear systems, such as limit cycles, which are oscillations that occur in the absence of external inputs.

1.1 Static Nonlinearities

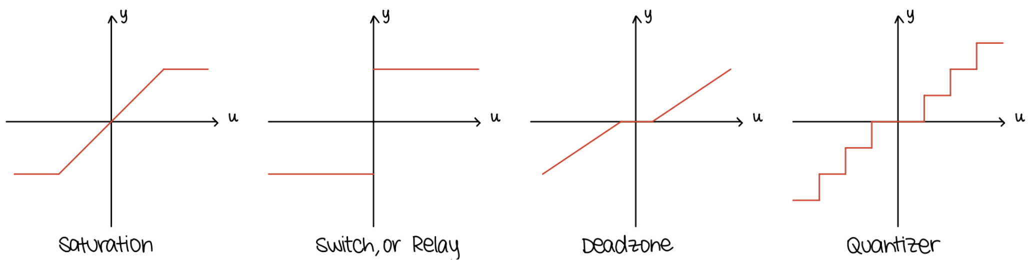
As a first step, let's focus on static nonlinearities.
As said before, in real world systems, many components can be nonlinear. We will mainly focus on those with a single static nonlinear feedback gain NL.

Let's consider the following feedback system:

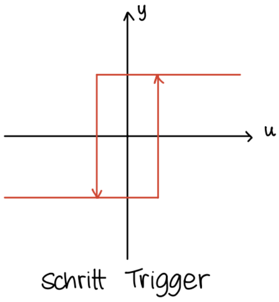


Here $L(s)$ is a linear time-invariant system and NL is a static nonlinearity.

NL can represent different types of nonlinearities such as static, memoryless ones:



But also dynamic nonlinearities with memory, such as:

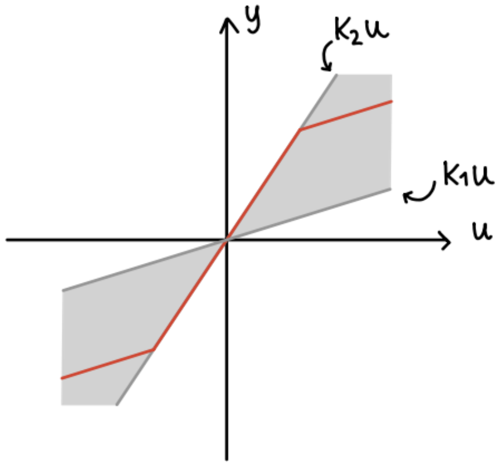


1.2 Absolute Stability

As we did with linear cases in the previous exercises, we want to assess stability of systems that include a nonlinear component.
To do so, we need to understand how the nonlinearity NL affects the loop gain of the system. We can do it by saying that NL gain is bounded within a sector $[k_1, k_2]$ meaning that

$$k_1 u \leq NL(u) \leq k_2 u \qquad \text{and} \qquad NL(0) = 0$$

This sector condition essentially bounds the "effective gain" of the nonlinearity between two constants.
We can also represent this graphically:

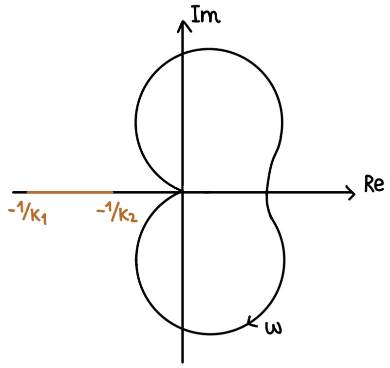


And say that the closed-loop system is absolutely stable in the sector $[k_1, k_2]$ if $u = 0$ is a globally asymptotically stable equilibrium point.

To analyze whether the closed-loop system is absolutely stable, we translate the sector condition into geometric constraints on the Nyquist plot of $L(s)$.
We can derive both necessary and sufficient conditions for absolute stability based on the Nyquist criterion.

Necessary Condition

A necessary condition for absolute stability follows directly from the sector bound definition. For the Nyquist criterion this means that the Nyquist plot of $L(s)$, instead of avoiding a single point, must avoid the entire line segment between $-1/k_1$ and $-1/k_2$ on the real axis and encircle it a number of times equal to the number of unstable poles of $L(s)$.

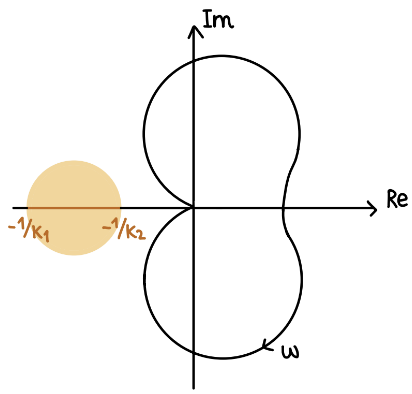


Note: Since it's a necessary condition, if it isn't fulfilled, absolute stability is not possible.

Sufficient Condition

The necessary condition for absolute stability states that the Nyquist plot of $L(s)$ must avoid the entire line segment between the points $-1/k_1$ and $-1/k_2$. However, avoiding only this segment guarantees stability only when the nonlinearity behaves as a constant gain within the sector $[k_1, k_2]$.
In practice, a sector-bounded nonlinearity may behave like a time-varying gain, taking any value within the sector. Therefore a stronger geometric constraint is required.

The *Circle Criterion* provides this stronger condition by enlarging the exclusion region from a line segment to the closed disk having the segment $[-1/k_1, -1/k_2]$ as its diameter. In other words, absolute stability is guaranteed if the Nyquist plot of $L(s)$ does not enter this disk and encircles it counterclockwise a number of times equal to the number of unstable poles of $L(s)$.



Note: This condition is sufficient but not necessary. If the Nyquist plot intersects the disk, the system may still be absolutely stable.

1.3 Describing Functions

In some cases, determining stability requires more advanced tools. One such tool is the Describing Function method, which approximates the behavior of a nonlinearity in the frequency domain.

Let's consider the effect of a sinsusoidal input

$$u(t) = A\sin(\omega t)$$

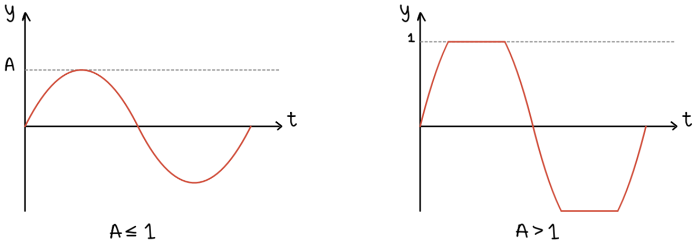
on a nonlinearity NL. The output $y(t)$ will now be of the form

$$y(t) = f(A\sin(\omega t))$$

This means that the output will generally be some periodic function with the same frequency ω as the input, but with the output now depending also on the amplitude A of the input signal.

Example: Saturation

$$y = \begin{cases} 1 & \text{if } u \geq 1 \\ u & \text{if } -1 < u < 1 \\ -1 & \text{if } u \leq -1 \end{cases}$$



With $u(t) = A\sin(\omega t)$

Since $y(t)$ is periodic, it can be expressed as a Fourier series:

$$y(t) = \frac{a_0}{2} + \sum_{i=1}^{\infty} [a_i \cos(i\omega t) + b_i \sin(i\omega t)]$$

Where the Fourier coefficients are given by:

$$a_i = \frac{1}{\pi} \int_{-\pi}^{pi} y(t)\cos(i\omega t) d(\omega t)$$

$$b_i = \frac{1}{\pi} \int_{-\pi}^{pi} y(t)\sin(i\omega t) d(\omega t)$$

We can further notice that for odd functions $y(t)$, all $a_i = 0$ and for even functions $y(t)$, all $b_i = 0$.
We can also observe that for many common nonlinearities, the higher harmonics (i.e., terms with $i > 1$) are often attenuated by the linear system. This allows us to approximate the output $y(t)$ by retaining only the first harmonic!

We can thus summarize everything as

$$N(A) = \frac{b_1(A)}{A} = \frac{1}{\pi A}\int_{-\pi}^{\pi} y(t)\sin(i\omega t)d(\omega t) \qquad \text{(for odd nonlinearities)}$$

$$N(A) = \frac{a_1(A)}{A} = \frac{1}{\pi A}\int_{-\pi}^{\pi} y(t)\cos(i\omega t)d(\omega t) \qquad \text{(for even nonlinearities)}$$

Where $N(A)$ is called the *Describing Function* of the nonlinearity NL. It is approximated as an amplitude-dependent gain.

We can further generalize the describing function to any input $u(t) = Ae^{j\omega t}$, leading to a complex describing function:

$$N(A,\omega) = \frac{c_1(A,\omega)}{A}e^{j\phi(A,\omega)}$$

Where $c_1(A,\omega)$ and $\phi(A,\omega)$ are respectively

$$c_1(A,\omega) = \sqrt{a_1^2 + b_1^2} \qquad \phi(A,\omega) = \arctan\left(\frac{a_1}{b_1}\right)$$

Example - Saturation

Let's consider again the saturation nonlinearity. The first thing we need to know is that the saturation nonlinearity is an odd function, so $a_i = 0$ for all i. Thus we only need to compute b_1.
Furthermore, we have two cases:

- if $A \leq 1$, the output is not saturated, therefore $b_1(A) = A$
- if $A > 1$, the output is saturated. In this case we define ψ as the value at which the signal is saturated:

$$A\sin(\psi) = 1 \implies \psi = \arcsin\left(\frac{1}{A}\right)$$

Thus we can compute b_1 as

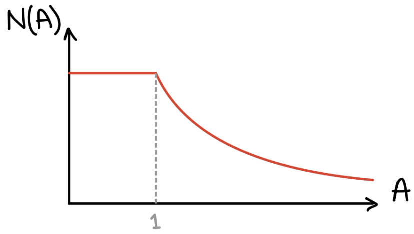
$$b_1 = \frac{4}{\pi}\int_0^{\pi/2} y(t)\sin(\psi)d\psi =$$

$$= \frac{4}{\pi}\int_0^{\psi} A\sin(\psi)\sin(\psi)d\psi + \frac{4}{\pi}\int_{\psi}^{\pi/2} 1\cdot\sin(\psi)d\psi = \cdots$$

$$= \frac{2}{\pi}A\left[\arcsin\left(\frac{1}{A}\right) + \frac{1}{A}\sqrt{1-\frac{1}{A^2}}\right]$$

Hence the describing function for the saturation nonlinearity is given by

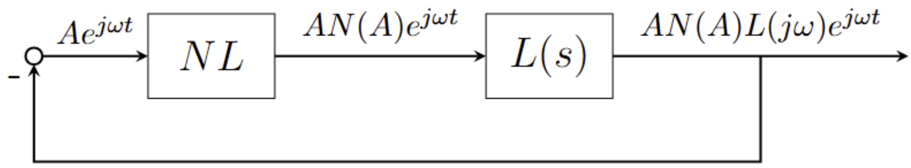
$$N(A) = \begin{cases} 1 & \text{if } A \leq 1 \\ \frac{b_1}{A} = \frac{2}{\pi}\left[\arcsin\left(\frac{1}{A}\right) + \frac{1}{A}\sqrt{1-\frac{1}{A^2}}\right] & \text{if } A > 1 \end{cases}$$



1.4 Limit Cycles

But how can we use these describing functions?

One of the most important applications is in analzying the existence and characteristics of limit cycles, which are self-sustaining oscillations that arise when the feedback loop stabilizes at a specific amplitude and frequency.
To see if a limit cycle is present, let's consider a feedback loop where the input of the NL is given by $Ae^{j\omega t}$



We can observe that if

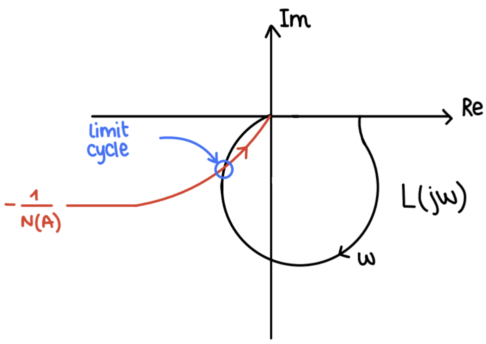
$$A = -AN(A)L(j\omega)$$

then the feedback loop is self-sustaining. The input $Ae^{j\omega t}$ produces an output of $-Ae^{j\omega t}$ which is negatively fed back into the system.
This will make the system to oscillate indefinetly causing a socalled limit cycle.

A condition for the existence of a limit cycle is given by the equation

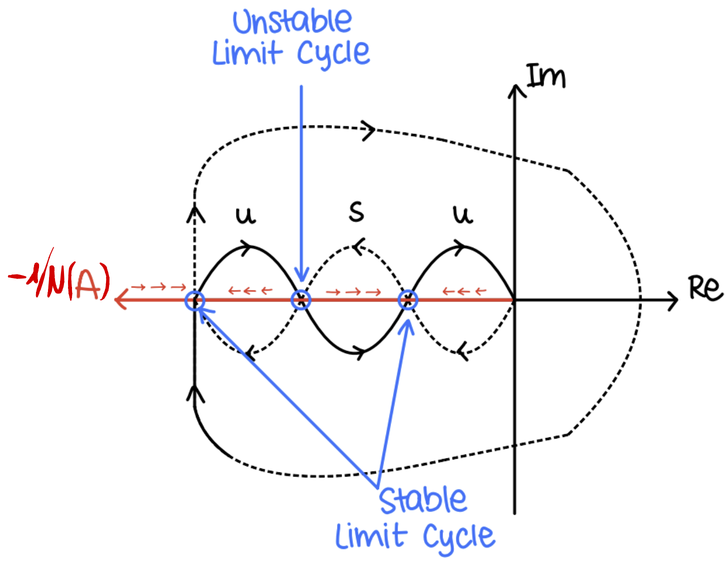
$$-\frac{1}{N(A)} = L(j\omega)$$

This equation can be solved graphically by plotting the Nyquist plot of $L(j\omega)$ and the locus of $-1/N(A)$ for varying A. The intersection points correspond to potential limit cycles, with the amplitude and frequency determined by the coordinates of the intersection.



Stability of Limit Cycles

To assess the stability of the limit cycles we can use the combination of the Nyquist plot and the plot of $-\frac{1}{N(A)}$.
We can say that if a point of $-1/N(A)$ is in an unstable region of the Nyquist plot of $L(j\omega)$, then the amplitude of oscillations will grow. On the other hand, if the point lies in a stable region, the amplitude will decrease.



Therefore we can conclude by saying that a limit cycle is stable if small perturbations in amplitude lead to corrections that bring the system back to the limit cycle.

Exam Problem - FS24

Problem: Consider the system in Figure 17, where L denotes a linear time-invariant system, and $N(u)$ a non-linear static gain.

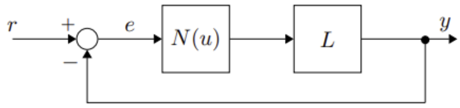


Figure 17: Feedback architecture, with linear time-invariant system L and non-linear static gain $N(u)$.

The static non-linear gain $N(u)$ is bounded by $[k_1, k_2]$, i.e,. $k_1 \leq N(u) \leq k_2, \forall u \in \mathbb{R}$, and the transfer function L is given by:

$$L(s) = \frac{5}{s^2 + 0.5s + 5}.$$

The Nyquist plot of L is shown in Figure 18 below.

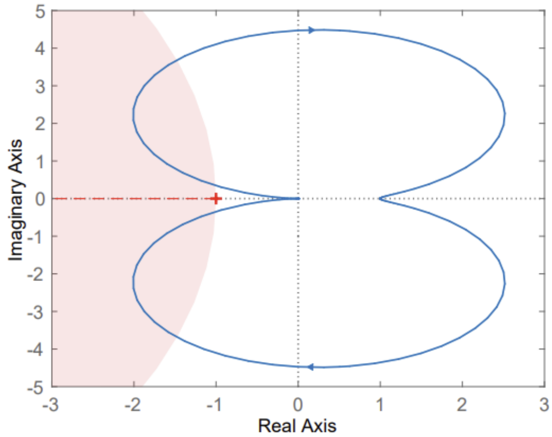


Figure 18: [Solid line] Nyquist plot of L. [Dashed line] plot of $N(u)$ as the gain varies within the segment $[\frac{-1}{k_1}, \frac{-1}{k_2}]$. [Shaded circle] Circle with diameter $[\frac{-1}{k_1}, \frac{-1}{k_2}]$.

Q39 **(1 Points)** Mark all correct statements.
Using the circle criterion, what can be concluded about the absolute stability of the closed-loop system?

- A The closed-loop system is absolutely stable.
- B The closed-loop system is not absolutely stable.
- C The absolute stability of the closed-loop system cannot be determined.

Exam Problem - HS23

Problem: Consider the closed-loop system T shown in Figure 23, where L is a linear time-invariant system and where NL is a saturation non-linearity.

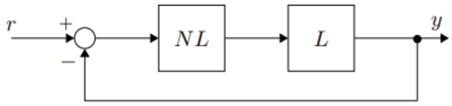


Figure 23: Standard feedback architecture. System L and saturation non-linearity denoted by NL.

The describing function $N(A)$ of the saturation non-linearity is given by,

$$N(A) = \frac{2}{\pi}\left[\arcsin\left(\frac{1}{A}\right) + \frac{1}{A}\sqrt{1-\left(\frac{1}{A}\right)^2}\right].$$

It is known that $L(s)$ has a pole at the origin, i.e. at $s = 0$, and that $L(s)$ has one unstable pole. Figure 24 shows the Nyquist plot of $L(s)$ (solid line) together with $\frac{-1}{N(A)}$ (dashed line). Further, assume that all assumptions required for a describing function analysis are met.

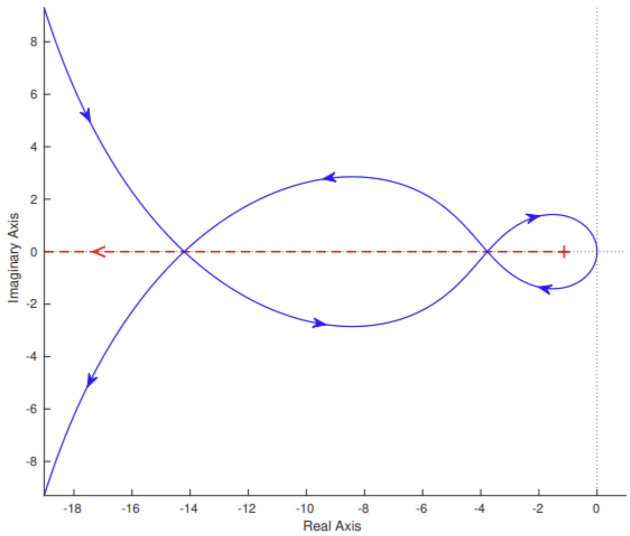


Figure 24: [Solid line] Nyquist plot of $L(s)$. [Dashed Line] Plot of $\frac{-1}{N(A)}$, where the arrow indicates the direction of $\frac{-1}{N(A)}$ as A increases.

Q40 (0.5 Points) Mark the correct answer.
First, without doing any stability analysis, what is the maximum number M of potential limit cycles that the closed-loop system T could support?

- [A] $M = 0$
- [B] $M = 2$
- [C] $M \to \infty$
- [D] $M = 1$

Q41 (1 Points) Mark the correct answer.
What is the number M_s of stable limit cycles that exist in the closed-loop system T?

- [A] $M_s = 0$
- [B] $M_s = 2$
- [C] $M_s \to \infty$
- [D] $M_s = 1$